1. Install python, conda, cellpose and the required dependecies

```
python -m pip install cellpose[gui]
```

Dependencies

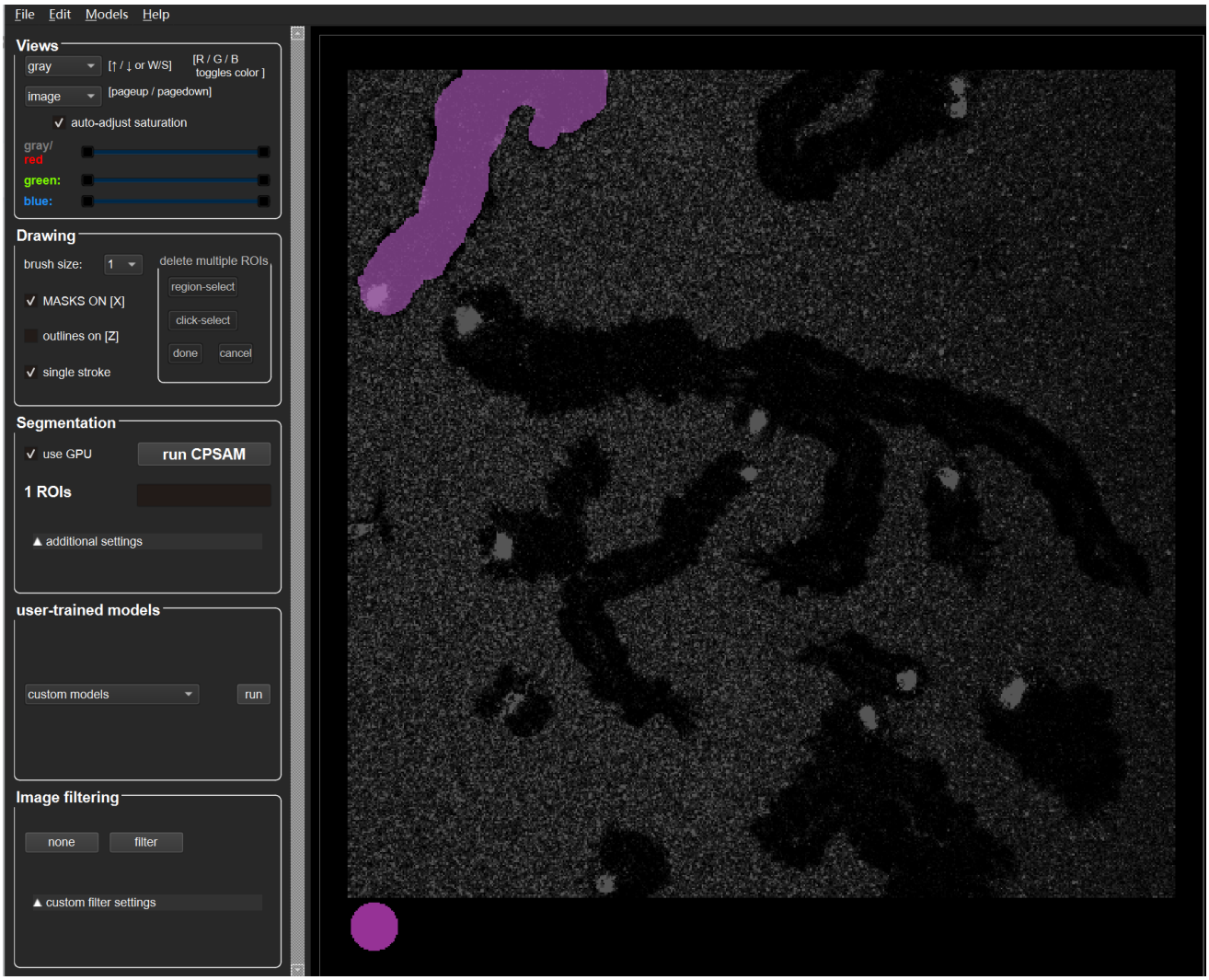
cellpose relies on the following excellent packages (which are automatically installed with conda/pip if missing):

- pytorch
- pyqtgraph
- PyQt6 or PySide
- numpy (>=1.20.0)
- scipy
- natsort
- tifffile
- imagecodecs
- roifile
- fastremap
- fill_voids

2. Manual annotate all the objects for more than 30 images

- **Separate the cell and track channels of czi files using the Step1_SpliChannelsByColor plugin in Fiji**
- **Open the track or cell image in cellpose GUI**

```
python -m pip install cellpose[gui]
```



- **Place the separated TIFF images and their corresponding labels into `CPSAM_v3/train`. To evaluate the training process, place a subset of the data into `CPSAM_v3/test` at an 2:8 ratio.**

3. Run the training process

```
python Train_CPSM.py `
    --train_dir train `
    --test_dir test `
    --pretrained_model "./pretrain_models/cpsam" `
    --model_name new_model `
    --n_epochs 150 `
    --batch_size 2 `
    --learning_rate 1e-5 `
    --weight_decay 0.00001 `
    --nimg_per_epoch 100
```